AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956
COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

# Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I058*

Team Name: *BUGLORDS*

Domain: *2 - Energy, Mobility and Smart Infrastructure*

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**Date:** 11th October 2026

# Presentation Outline

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# Problem Statement

AI-Based Priority-Aware Electricity Allocation During Power Shortages.

# Problem Understanding

- Everyone connected to the Electricity feels it, but the people hurt most are those who depend on electricity for safety: patients, people on medical equipment, and emergency services.
- There are two causes. First, supply sometimes falls short of demand. Second, the way shortages are handled is neither need based nor prioritized.
- When power is cut blindly i.e. cutting power without checking who needs it most. Critical services and vulnerable people face safety risks. Our system reduces that harm by protecting the most critical users first

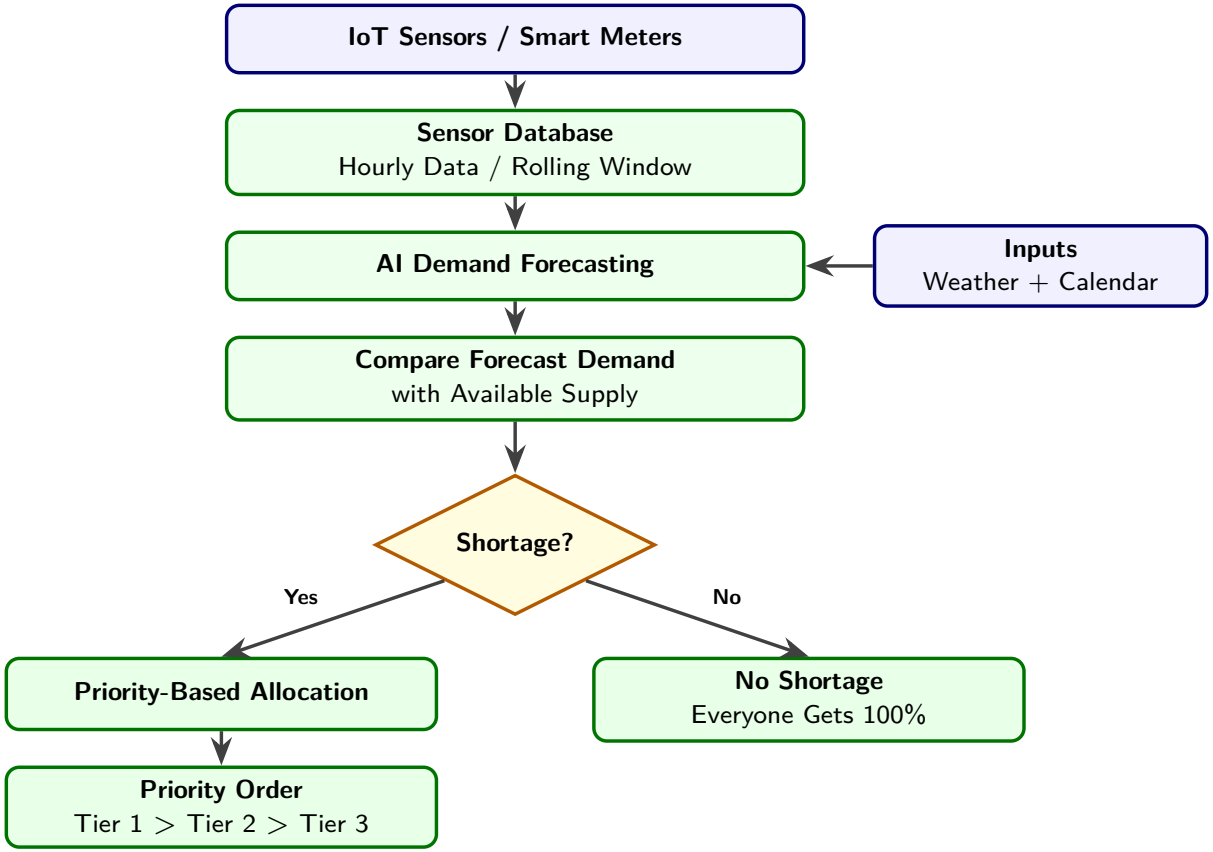
# Existing Gaps & Challenges

- Power cuts today don't consider who needs electricity the most.
- Missing a system that looks ahead and decides who must keep power first
- No system today predicts how much power people need and protects hospitals and emergency services first

# Our Proposed Solution

- When there isn't enough electricity, our system predicts who needs it and gives it to the most critical users first
- AI predicts how much power each type of user will need, then shares the limited supply by priority, and corrects the plan as actual use changes and Hospitals and water supply stay powered, and the rest of the shortage is shared fairly.
- Our system makes sure that critical services such as hospitals, water supply and emergency services stay powered, even when there isn't enough electricity for everyone. The remaining supply is shared fairly among homes, shops and industries, with every group guaranteed a minimum share instead of being left in the dark. Because every decision is planned in advance and explained in plain words, users face fewer surprises, and utility operators can make faster, clearer and more trusted decisions.

# Technical Approach




**Tech Stack**

**Frontend:** Streamlit

**Backend:** SQLite

**Database:** Python (pandas, NumPy)

**AI/ML, APIs, or Hardware:** Forecast model, Claude API, ESP32 (planned)

# Innovation & Uniqueness

- Instead of cutting power by fixed schedules or whole areas, our system forecasts how much power each type of user will need and shares the limited supply using priority bands. Every tier gets a guaranteed minimum and a maximum share, so critical users are protected and no one is cut to zero. An AI assistant also explains each decision in plain words.

- The problem is not only that power runs short, but that the shortage is handled late and without regard to importance. Forecasting from smart-meter data lets us plan before the shortage begins. Priority bands turn the utility's policy into clear, adjustable rules. Anomaly detection and a what-if simulator help operators trust the data and test scenarios, and a human can always override the system.

- Most existing methods do only one part: fixed rotation, a few dedicated priority feeders, or costly backup generators. Our key difference is that forecasting, priority-based allocation and clear explanations work together in one system. This lets us protect critical services first and share the rest of the shortage fairly, with every decision explained.

# Feasibility & Scalability

- **Feasibility:** Uses existing IoT meters + open-source AI; prototype-ready with no new hardware.
- **Scalability:** Campus $\rightarrow$ city grid by adding sensors and cloud capacity; the tier logic stays the same.
- **Risks:** Data gaps and forecast errors, mitigated by fallback averages, frequent re-forecasting and a reserve margin.

# Use Cases

- Grid operator handling an evening shortage :
  At 6 PM, a utility's grid operator expects a shortage. Demand is forecast at 2000 kW (Tier 1: 1000 kW, Tier 2: 500 kW, Tier 3: 500 kW), but only 1500 kW of supply is available. Instead of switching off areas blindly, the system applies the priority bands: Tier 1 gets at least 90 percentage of its demand, Tier 2 gets 40-90 percentage and Tier 3 gets 40-60 percentage. Each tier first receives its minimum (900, 200 and 200 kW), and the spare 200 kW goes to the higher tiers first. The result is 1000 kW for hospitals and water supply (100 percentage), 300 kW for homes and schools (60 percentage), and 200 kW for shops and factories (40 percentage). The operator sees the plan, reads the AI explanation of why each tier got its share, and approves it or overrides it.
  Note: Additional slides may be included only for this section.

# Use Cases

- Hospital and households with medical equipment during a heatwave:
  During a heatwave, air-conditioner use pushes demand up and supply falls short. A hospital running ICUs and a household using an oxygen concentrator cannot safely lose power. The system forecasts the higher demand from the temperature, places critical users in Tier 1, and reserves their supply before any other group is reduced. Because the plan is made in advance, the hospital and the household receive an early alert and can prepare, instead of facing a sudden cut.
  Note: Additional slides may be included only for this section.

# Use Cases

- Utility planner testing scenarios and catching faults:
  A utility planner wants to prepare for a possible generation drop tomorrow. Using the what-if simulator, the planner lowers the supply by 20 percentage and sees how much each tier would receive, and at what point emergency mode starts. At the same time, anomaly detection flags a sudden spike in one area's smart-meter readings, which may point to a fault or theft, so the planner can check it before it distorts the forecast. This helps the utility plan early, avoid surprises and explain its decisions to the public.
  Note: Additional slides may be included only for this section.

# Impacts and Benefits

- Hospitals, water supply and emergency services keep their power first, so patients and essential services are not put at risk.Utility operators get forecasts, what-if testing and clear explanations that make their decisions faster and easier to trust.
- Socially, the system protects lives and public safety by keeping critical services running and by sharing the shortage fairly, which builds public trust in the utility. Economically, planned and predictable cuts reduce the losses that shops and industries face from sudden outages, and they reduce how much people spend on backup generators. Environmentally, fewer emergency generators running on diesel means lower fuel use and pollution during shortages.
- We will measure success by three things. First, critical users (Tier 1) always receive at least 90 percentage of their forecast demand, and in our 6 PM example they receive 100 percentage (1000 kW of 1000 kW). Second, forecast accuracy, tracked as the average percentage error (MAPE), which is about 2 percentage on our simulated demo data and will be re-measured on real smart-meter data.

# Sustainable Development Goals (SDGs)

| SDG Number | Target Number | Goal Name | Connection |
|---|---|---|---|
| **SDG 7** | Target 7.1 | Affordable and Clean Energy | Makes electricity supply more reliable by protecting critical users during shortages and sharing limited power fairly. |
| **SDG 9** | Target 9.1 | Industry, Innovation and Infrastructure | Uses AI and smart-meter data to make the power grid smarter and more resilient. |

# Individual Member Contribution

| Member | Contribution or responsibility |
|---|---|
| **[Member 1 DHARUNEASH-WAR E ]** | Problem statement Ideations ,technical tools and architecture |
| **[Member 2 S M SRI-HARIPRASATH ]** | Research paper analysis and Feature Description |
| **SHABHARISH J M** | Presentation, poster and documentation |
| **RUPESH S** | Presentation, poster and documentation |

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